

MOHD ANAS

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EDUCATION

B.Tech, Information Technology

IIIT Vadodara, Gujarat, India

Expected Graduation: Jan 2027

7.39 / 10.0 GPA

Coursework: Computer Networking, Operating Systems, DBMS, Data Structures & Algorithms, Machine Learning

PROFESSIONAL EXPERIENCE

Aarogya Kaya, Remote: SDE Intern

July 2025 - Jan 2026

- Architected **scalable backend microservices** in Go for seamless third-party integrations (Shiprocket, Instagram Graph API), implementing **CQRS** architecture to decouple data synchronization commands from read-intensive client queries.
- Applied **SOLID principles** and **structural design patterns** (Dependency Injection, Strategy) to translate business requirements into scalable systems, reducing technical debt and improving maintainability.
- Engineered secure, high-throughput API pipelines for social media integrations, optimizing request payloads to achieve **sub-250 ms response times** while enforcing data security.
- Improved deployment and frontend performance via CI/CD pipelines enforcing unit/integration testing and Lighthouse audits, reducing release rollback incidents to **near-zero** and maintaining **85+ Core Web Vitals scores**.

Next.js (Vercel), Remote: Open Source Contributor

July 2025 - Present

- Ported the patch-next utility from CommonJS to TypeScript with ESM, improving CI testing workflows and reducing iteration time for Next.js contributors.

PROJECTS

a3.store - Python, gRPC, Kubernetes (EKS), AWS, SQLite, Prometheus

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Distributed Key-Value Database

- Built and operated a **leaderless distributed key-value store**, handling node joins, failures, and data replication using **gossip-based membership** and **anti-entropy repair** across a 3-node cluster.
- Deployed the system on **AWS EKS** using Kubernetes StatefulSets and persistent volumes, managing rolling upgrades, pod restarts, and **self-healing behavior**.
- Built **operational dashboards** and **disaster-recovery workflows** to make the system debuggable, reliable, and usable in production-like environments.

IntoxAI - Python, LangChain, FastAPI, RAG

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Retrieval-Augmented Generation (RAG) system for xAI FAQs with modular LLM orchestration and retrieval pipeline.

- Architected a modular RAG pipeline using **Factory** and **Strategy** design patterns for pluggable embedding models, retrievers, and LLM providers.
- Integrated **LangChain**, **Google AI APIs**, and **ChromaDB** to implement semantic search, context-aware retrieval, and low-latency embedding-based document querying.
- Backend services with token-efficient prompt handling using **tiktoken**, optimizing Gemini API usage and latency.

TECHNICAL SKILLS

Programming Languages: Python, Go, C/C++, JavaScript, TypeScript, Rust, SQL

Frameworks & Libraries: Node.js, React, Next.js, gRPC, WebSockets (Socket.io), CLI Tooling

Tools & Technologies: Git, GitHub, Docker, **Kubernetes (EKS)**, **AWS** (EC2, EKS, IAM, VPC), Terraform, **Linux**, Jenkins, Redis, SQLite, Observability (Prometheus, Grafana, Loki)

AI & LLMs: LangChain, HuggingFace, TensorFlow, PyTorch, Neural Network, DeepLearning, NLP, Generative AI

CORE COMPETENCIES

Backend Engineering, Distributed Systems, DevOps, API Integrations, Automation, Product Engineering

LEADERSHIP & ACTIVITIES

Technical Committee, IIIT Vadodara - Core Member

March 2024 - Present

- Organized an intra-college hackathon and mentored **25+ teams**.